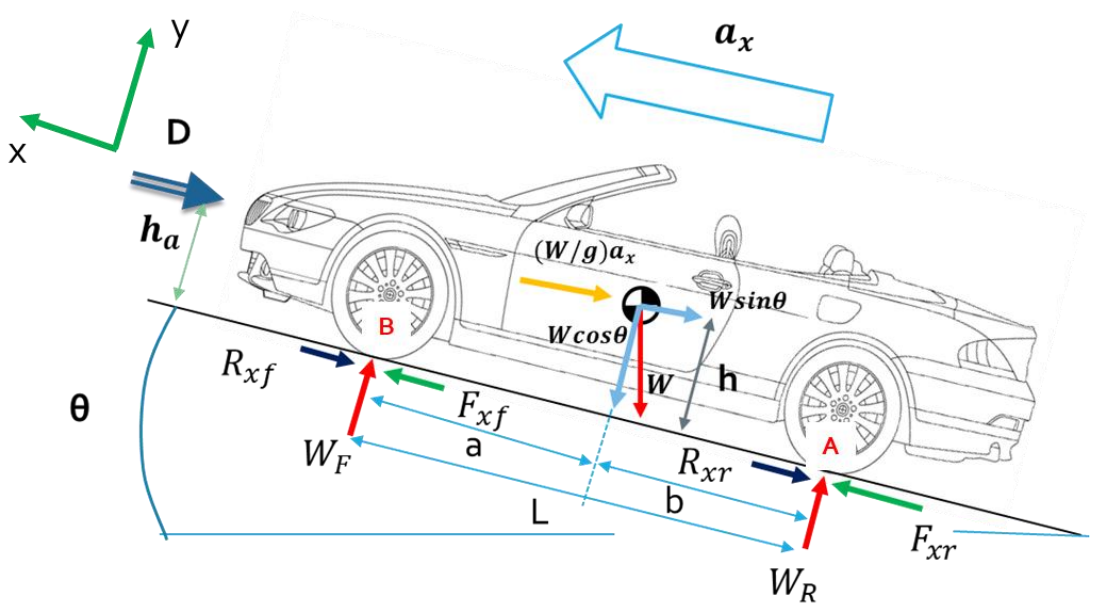
Longitudinal dynamics - Dynamic Force analysis of a Vehicle in Acceleration - Problem

A rear wheel car is driven by a traction force of 2000 N , what will be the acceleration of the car if the car is starting from 5 m/s(18 kmph) velocity , drag effects are negligible and Car is running on a road inclined at 2 degree .
Rolling resistance coefficient is 0.0150
Vehicle weight is 2000 kg
wheelbase is 2500 mm
and CG is located 1.4 m aft (rear) of front wheel
Height of CG from road is 500mm
How much time will it take for vehicle to reach 36 kmph



$$F_{xf} + F_{xr} = \left(+\left(\frac{W}{g}\right)(a_x)\right) + \boldsymbol{R_{xf}} + \boldsymbol{R_{xr}} + \boldsymbol{W\sin\theta} + \boldsymbol{D}$$

$\theta = 2°$ $\quad D = 0$ $\quad F_{xf} = 0$ $\quad C_{rr} = 0.0150$

$a = 1.4\ m; b = 1.1\ m$ $\quad F_{xr} = 500\ N$ $\quad W = 2000 * 9.81\ N = 19620\ N$

$$\boldsymbol{R_{xf}} = \boldsymbol{C_{rr}W_F}$$

$$\boldsymbol{R_{xr}} = \boldsymbol{C_{rr}W_R}$$

$$W_r = \frac{+Dh_A + \left(\frac{W}{g}a_x h\right) + W\sin\theta h + W\cos\theta a}{L}$$

$$W_r = \frac{+(2000 * a_x * 0.5) + 19620\ \sin 2\ * 0.5 + 19620\ \cos 2\ * 1.4}{2.5}$$

$$W_r = 400a_x + 11117.45$$

$$2000 = \left(+\left(\frac{19620}{9.81}\right)(a_x)\right) + (C_{rr}W_F) + (C_{rr}W_R) + 19620\ \sin 2° + 0$$

$$2000 = \left(+\left(\frac{19620}{9.81}\right)(a_x)\right) + (0.0150\ * W_F) + (0.0150 * W_R) + 684.728$$

$$2000 = \left(+\left(\frac{19620}{9.81}\right)(a_x)\right) + (0.0150\ * (-400a_x + 10638.14)) + (0.0150 * (400a_x + 11117.45)) + 684.728$$

$$2000 = 2000a_x - 6a_x + 159.57 + 6a_x + 166.76 + 684.728$$

$$\boxed{a_x = 0.49\ m/s^2}$$

$$W_f = \frac{-Dh_A - \left(\frac{W}{g}a_x h\right) - W\sin\theta h + W\cos\theta b}{L}$$

$$W_f = \frac{-(2000 * a_x * 0.5) - 19620\ \sin 2\ * 0.5 + 19620\ \cos 2 * 1.1}{2.5}$$

$$W_f = -400a_x + 10638.14$$

$\boxed{a_x = 0.49\ m/s^2}$

$$a = \frac{v - u}{t}$$

u = 5 m/s ; v = 10 m/s (36 kmph)

$$0.49 = \frac{10 - 5}{t}$$

$$\boldsymbol{t = 10\ seconds}$$